

# **Design activities**

## **Session I**

### 1. Topology

- Selecting and placing devices
- Interconnecting devices

### 2. Providing connectivity between directly connected devices

- Configuring addresses for servers and hosts
- Configuring DHCP servers
- Configuring network interface cards

### 3. Testing

### 4. Technical report (1-2 pages)

## **Session II**

### 1. Configuring VTP, VLAN and STP at layer 2 equipments

### 2. Configuring routes and routing protocols at layer 3 equipments

### 3. Testing

### 4. Technical report (1-2 pages)

## **Session III**

### 1. Configuring NAT on the routers connected to Internet

### 2. Testing

### 3. Technical report (1-2 pages)

## **Session IV**

1. Configuring HTTP, FTP, DNS and MAIL servers
2. Testing
3. Technical report (1-2 pages)

## **Session V**

1. Securing the network
2. Testing
3. Technical report (1-2 pages)

## **Session VI**

1. Test scenarios
2. Technical report (1-2 pages)

## **Session VII**

10. Final presentation

## **Project topics**

### **Theme 1:**

It is considered a commercial building with 3 levels. Use network address 172.27.0.0/16 for intranet network, network address 210.2.2.64/27 for DMZ and network address 210.2.2.32/27 for access to the exterior. Design four VLANs (one for each floor and one for traffic management). VTP protocol will be used to configure VLANs. Redundancy will be provided through network cabling and configuration. Host addresses will be assigned dynamically using DHCP servers. The minimum number of users served by each VLAN is 200. HTTP, FTP, DNS and Mail servers will be placed in the DMZ and will have public addresses. Web domain name will include the student's name. To ensure connectivity static routes will be configured. Access to the exterior will be done using NAT on the router that controls the DMZ, on the following range of public addresses: 210.2.2.35-210.2.2.62.

Connecting to the ISP will be achieved through an Ethernet interface having the address 210.2.2.34/27. ISP address is 210.2.2.33/27. Internet network will be simulated through a server and a computer.

For securing network devices the following configurations will be performed: users on different privilege levels will be defined, passwords will be encrypted, remote configuration will be done only through ssh, VTP protocol will be secured.

Two additional security measures will be presented and implemented.

## Theme 2:

It is considered a commercial building with 3 levels. Use network address 172.27.0.0/16 for intranet network, network address 210.2.2.64/27 for DMZ and network address 210.2.2.32/27 for access to the exterior. Design four VLANs (one for each floor and one for traffic management). VTP protocol will be used to configure VLANs. Redundancy will be provided through network cabling and configuration. Host addresses will be assigned dynamically using a single DHCP server located in the VLAN corresponding to the first floor. The minimum number of users served by each VLAN is 200. HTTP, FTP, DNS and Mail servers will be placed in the DMZ and will have public addresses. Web domain name will include the student's name. To ensure connectivity static routes will be configured. Access to the exterior will be done using NAT on the router that controls the DMZ, on the following range of public addresses: 210.2.2.35-210.2.2.62.

Connecting to the ISP will be achieved through an Ethernet interface having the address 210.2.2.34/27. ISP address is 210.2.2.33/27. Internet network will be simulated through a server and a computer.

For securing network devices the following configurations will be performed: users on different privilege levels will be defined, passwords will be encrypted, remote configuration will be done only through ssh, VTP protocol will be secured.

Two additional security measures will be presented and implemented.

## Theme 5:

It is considered a commercial institution with 3 buildings. Use network address 172.27.0.0/16 for intranet network, network address 210.2.2.64/27 for DMZ and network address 210.2.2.32/27 for access to the exterior. There will be designed 3 subnets for users (one for each building). The users will have the possibility to connect to the network both by cable and wireless. Redundancy will be provided through network cabling and configuration. Host addresses will be assigned dynamically using DHCP servers configured at the router level. The minimum number of users served by each subnet is 200. HTTP, FTP, DNS and Mail servers will be placed in the DMZ and will have public addresses. Web domain name will include the student's name. The routing will be done using the OSPF protocol for which security options will be implemented. Access to the exterior will be done using NAT on the router that controls the DMZ, on the following range of public addresses: 210.2.2.35-210.2.2.62.

Connecting to the ISP will be achieved through an Ethernet interface having the address 210.2.2.34/27. ISP address is 210.2.2.33/27. Internet network will be simulated through a server and a computer.

For securing network devices the following configurations will be performed: users on different privilege levels will be defined, passwords will be encrypted, remote configuration will be done only through ssh, wireless networks will be secured with WPA2.

Two additional security measures will be presented and implemented.